

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course:T.Y.B.TECH **Branch :** Mechanical Eng./Mechanical Eng .(sandwich) **Semester :** VI

Subject Code & Name: BTME603_Y 19 & Applied Thermodynamics-II

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The air standard efficiency of Diesel cycle compared to Otto cycle for given compression ratio is a. same b. less c. more d. None	CO1	1
2	The compression ratio of diesel engine usually in the range of a.1-5 b. 6-12 c.15-25 d.21-30	CO1	1
3	The reason of supercharging in any engine a. increase efficiency b. increase power c. reduced weight & bulk for given output d. efficiency fuel economy	CO2	1
4	Which of the following major exhaust emission from CI engine Compared to SI engines? a. Oxides of nitrogen B. particulates C. Carbon monoxide and carbon dioxide d. unburnt hydrocarbon	CO2	1
5	In Vapour absorption refrigeration system, the lowest temperature during the cycle is observed after a.condensation b. expansion c. evaporation d. compression	CO3	1
6	Vapour compression refrigeration system uses as input energy a. electricity energy b. water energy c. low grade heat energy D. non-conventional energy only	CO3	1

7	For summer air conditioning system, the relative humidity should not more than				CO4	1
	A 60%	b.75%	c. 40%	d. 90%		
8	Specific humidity is also called				CO4	1
	a. relative humidity	b. humidity ratio	c. absolute humidity	d. saturation humidity		
9	Which of the statement is INCORRECT with respect to reheat cycle				CO4	1
	a. Steam rate increase	b. Turbine work increase	c. Net Work done increase	d. work ratio increase		
10	What is use of surge tank in hydroelectric power plant				CO5	1
	a. controlling water flow rata	b. changing water flow rate	c. counteracts sound of water	d. controlling pressure changes in the system		
11	Which is greenhouse effect				CO5	1
	A, O ₂	b.CO	c. CO ₂	d. N ₂		
12	Which of the following is limitation gas turbine power Plant				CO5	1
	a. Not self - starting	b. Higher rotor speed	c. low efficiency at part load	d. All of the above		
Q. 2	Solve the following.					12
A)	Draw and Explain Otto cycle with P-V & T-S diagram				CO1	6
B)	Write the difference between S.I and C.I Engine				CO1	6
Q.3	Solve the following.					12
A)	Draw circuit diagram battery or coil ignition system and write function of each component				CO2	6
B)	Explain the various pollutants from SI & CI engine and their controls				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Draw and explain ideal vapour- Absorption refrigeration cycle on p-h & T-s diagram				CO3	6
B)	Explain in brief Lithium bromide-water system				CO3	6
C)	Write the difference between Vapour Absorption and vapour Compression system.				CO3	6

Q.5	Solve Any Two of the following.		12
A)	Explain the term SHF, RSHF& By pass factor in brief.	CO4	6
B)	Define the following: Relative Humidity, Specific Humidity and Room Sensible heat factor	CO4	6
C)	Differentiate between horizontal axis wind turbine and vertical axis Wind turbine	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Describe the general layout of hydroelectric power plant and explain function of each component	CO5	6
B)	Describe Gas Turbine power plant in brief.	CO5	6
C)	Explain in brief safety requirement of nuclear power plant	CO5	6
	*** End ***		